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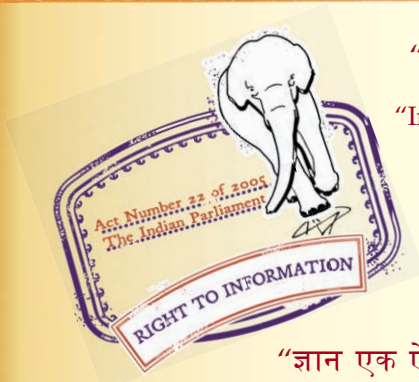
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IS 1570-5 (1985): Schedules for Wrought Steels, Part 5: Stainless and Heat-resisting Steels [MTD 16: Alloy Steels and Forgings]



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“Knowledge is such a treasure which cannot be stolen”



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# *Indian Standard*

## SCHEDULES FOR WROUGHT STEELS PART 5 STAINLESS AND HEAT-RESISTING STEELS

*( Second Revision )*

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**BUREAU OF INDIAN STANDARDS**  
MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG  
NEW DELHI 110002

# Indian Standard

## SCHEDULES FOR WROUGHT STEELS PART 5 STAINLESS AND HEAT-RESISTING STEELS

### ( Second Revision )

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## *Indian Standard*

### SCHEDULES FOR WROUGHT STEELS

#### PART 5 STAINLESS AND HEAT-RESISTING STEELS

### *( Second Revision )*

#### 0. FOREWORD

**0.1** This Indian Standard ( Part 5 ) ( Second Revision ) was adopted by the Indian Standards Institution on 30 October 1985, after the draft finalized by the Alloy Steels and Special Steels Sectional Committee had been approved by the Structural and Metals Division Council.

**0.2** This standard was first published in 1961 and subsequently revised in 1972. On the basis of experience gained in the production and use of these steels, it has been decided to issue the present revision of this standard.

**0.3** The following major modifications have been incorporated in this revision:

- a) Chemical compositions of some of the grades in Table 1 have been modified in the light of the indigenous production of these steels.
- b) Tables 6, 11, 12 and 14 of the first revision have been deleted, as sufficient data is not available and such products are not being produced on a mass scale in the country.
- c) Other tables have been rationalized and modified on the basis of indigenous production of the steels.

**0.4** This schedule is not intended to be used as a standard and for detailed requirements, reference should be made to the following standards:

- a) IS : 5522-1978 Stainless steel sheets and coils (*first revision*)
- b) IS : 6527-1972 Stainless steel wire rod
- c) IS : 6528-1972 Stainless steel wire
- d) IS : 6529-1972 Stainless steel blooms, billets and slabs for forgings
- e) IS : 6603-1972 Stainless steel bars and flats

- f) IS : 6911-1972 Stainless steel plate, sheet and strip
- g) IS : 7494-1981 Steel for valves for internal combustion engines  
( *first revision* )
- h) IS : 9294-1979 Cold rolling stainless steel strips for razor blades
- j) IS : 9516-1980 Heat resisting steels

**0.5** Mechanical properties given in Tables 2 to 6 are applicable to heat treated steels according to the stipulations made in Table 7. The specified mechanical properties are applicable to test samples taken from a standard location which should be included in the specification.

**0.6** This standard ( Part 5 ) deals with stainless and heat-resisting steels, other parts of IS : 1570 have already been published are:

- IS : 1570 ( Part 1 )-1978 Schedules for wrought steels: Part 1 Steels specified by tensile and/or yield properties  
( *first revision* ).
- IS : 1570 ( Part 2 )-1979 Schedules for wrought steels: Part 2 Carbon steels ( unalloyed steels ) ( *first revision* ).
- IS : 1570 ( Part 3 )-1979 Schedules for wrought steels: Part 3 Carbon and carbon manganese free cutting steels  
( *first revision* ).

**0.7** Schedules IV and VI of IS : 1570 are also under revision and these will also be printed as separate standards in due course.

**0.8** Information on the assessment of the ruling sections of bars, forgings and parts is given in Appendix A.

**0.9** For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS : 2-1960\*. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

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## **1. SCOPE**

**1.1** This standard ( Part 5 ) is applicable to steels with a high alloy content primarily intended for corrosion and heat-resisting purposes and supplied to specified chemical composition as given in Table 1. The mechanical properties of these steels are given in Tables 2 to 6.

\*Rules for rounding off numerical values ( *revised* ).



TABLE 1 SPECIFIED CHEMICAL COMPOSITIONS OF HIGH ALLOY STEELS, STAINLESS STEELS AND HEAT-RESISTING STEELS

| SL<br>No. | STEEL<br>DESIGNATION | ( Clause 1.1 )<br>CHEMICAL COMPOSITION |           |           |           |           |          |           |          |       |       | REMARKS    |
|-----------|----------------------|----------------------------------------|-----------|-----------|-----------|-----------|----------|-----------|----------|-------|-------|------------|
|           |                      | C                                      | Si        | Mn        | Ni        | Cr        | Mo       | Ti        | Nb       | S Max | P Max |            |
| (1)       | (2)                  | (3)                                    | (4)       | (5)       | (6)       | (7)       | (8)      | (9)       | (10)     | (11)  | (12)  | (13)       |
| 1.        | *X04Cr12             | 0.08 Max                               | 1.0 Max   | 1.0 Max   | —         | 11.5/13.5 | —        | —         | —        | 0.030 | 0.040 |            |
| 2.        | *X12Cr12             | 0.08, 0.15                             | 1.0 Max   | 1.0 Max   | 1.0 Max   | 11.5/13.5 | —        | —         | —        | 0.030 | 0.040 |            |
| 3.        | X20Cr13              | 0.16/0.25                              | 1.0 Max   | 1.0 Max   | 1.0 Max   | 12.0/14.0 | —        | —         | —        | 0.030 | 0.040 |            |
| 4.        | X30Cr13              | 0.26/0.35                              | 1.0 Max   | 1.0 Max   | 1.0 Max   | 12.0/14.0 | —        | —         | —        | 0.030 | 0.040 |            |
| 5.        | X40Cr13              | 0.35/0.45                              | 1.0 Max   | 1.0 Max   | 1.0 Max   | 12.0/14.0 | —        | —         | —        | 0.030 | 0.040 |            |
| 6.        | X07Cr17              | 0.12 Max                               | 1.0 Max   | 1.0 Max   | 0.5 Max   | 16.0/18.0 | —        | —         | —        | 0.030 | 0.040 |            |
| 7.        | X15Cr16Ni2           | 0.10/0.20                              | 1.0 Max   | 1.0 Max   | 1.25/2.50 | 15.0/17.0 | —        | —         | —        | 0.030 | 0.045 |            |
| 8.        | X108Cr17Mo           | 0.95/1.20                              | 1.0 Max   | 1.0 Max   | 0.50 Max  | 16.0/18.0 | 0.75 Max | —         | —        | 0.030 | 0.045 |            |
| 9.        | X02Cr19Ni10          | 0.030 Max                              | 1.0 Max   | 2.0 Max   | 8.0/12.0  | 17.5/20.0 | —        | —         | —        | 0.030 | 0.045 |            |
| 10.       | X04Cr19Ni9           | 0.08 Max                               | 1.0 Max   | 2.0 Max   | 8.0/10.5  | 17.5/20.0 | —        | —         | —        | 0.030 | 0.045 |            |
| 11.       | X07Cr18Ni9           | 0.15 Max                               | 1.0 Max   | 2.0 Max   | 8.0/10.0  | 17.0/19.0 | —        | —         | —        | 0.030 | 0.045 |            |
| 12.       | X04Cr18Ni10Ti        | 0.08 Max                               | 1.0 Max   | 2.0 Max   | 9.0/12.0  | 17.0/19.0 | —        | 5XC-0.80  | —        | 0.030 | 0.045 |            |
| 13.       | X04Cr18Ni10Nb        | 0.08 Max                               | 1.0 Max   | 2.0 Max   | 9.0/12.0  | 17.0/19.0 | —        | —         | 10XC-1.0 | 0.030 | 0.045 |            |
| 14.       | X04Cr17Ni12Mo2       | 0.08 Max                               | 1.0 Max   | 2.0 Max   | 10.0/14.0 | 16.0/18.0 | 2.0/3.0  | —         | —        | 0.030 | 0.045 |            |
| 15.       | X02Cr17Ni12Mo2       | 0.030 Max                              | 1.0 Max   | 2.0 Max   | 10.0/14.0 | 16.0/18.0 | 2.0/3.0  | —         | —        | 0.030 | 0.045 |            |
| 16.       | X04Cr17Ni12Mo2Ti     | 0.080 Max                              | 1.0 Max   | 2.0 Max   | 10.0/14.0 | 16.0/18.0 | 2.0/3.0  | 5XC-0.80† | —        | 0.030 | 0.045 |            |
| 17.       | X04Cr19Ni13Mo3       | 0.08 Max                               | 1.0 Max   | 2.0 Max   | 11.0/15.0 | 18.0/20.0 | 3.0/4.0  | —         | —        | 0.030 | 0.045 |            |
| 18.       | X15Cr25N             | 0.20 Max                               | 1.0 Max   | 1.5 Max   | —         | 23.0/27.0 | —        | —         | —        | 0.030 | 0.045 | N 0.25 Max |
| 19.       | X07Cr17Mn12Ni4       | 0.12 Max                               | 1.0 Max   | 10.0/14.0 | 3.5/5.5   | 16.0/18.0 | —        | —         | —        | 0.030 | 0.045 | N 0.25 Max |
| 20.       | X10Cr17Mn6Ni4        | 0.20 Max                               | 1.0 Max   | 4.0/8.0   | 3.5/5.5   | 16.0/18.0 | —        | —         | —        | 0.030 | 0.045 |            |
| 21.       | X15Cr24Ni13          | 0.20 Max                               | 1.5 Max   | 2.0 Max   | 11.0/15.0 | 22.0/25.0 | —        | —         | —        | 0.030 | 0.045 |            |
| 22.       | X40Ni14Cr14W3Si2     | 0.35/0.50                              | 2.0 Max   | 1.0 Max   | 12.0/15.0 | 12.0/15.0 | —        | —         | —        | 0.035 | 0.045 | W 2.0/3.0  |
| 23.       | X20Cr25Ni20          | 0.25 Max                               | 2.5 Max   | 2.0 Max   | 18.0/21.0 | 24.0/26.0 | —        | —         | —        | 0.030 | 0.045 |            |
| 24.       | X04Cr25Ni20          | 0.08 Max                               | 1.5 Max   | 2.0 Max   | 19.0/22.0 | 24.0/26.0 | —        | —         | —        | 0.030 | 0.045 |            |
| 25.       | X45Cr9Si3            | 0.040/0.50                             | 3.0/3.75  | 0.30/0.60 | 0.50 Max  | 7.5/9.5   | —        | —         | —        | 0.050 | 0.045 |            |
| 26.       | X80Cr20Si2Ni1        | 0.75/0.85                              | 1.75/2.25 | 0.20/0.60 | 1.20/1.70 | 19.0/21.0 | —        | —         | —        | 0.030 | 0.045 |            |
| 27.       | X66Cr13              | 0.60/0.72                              | 0.5 Max   | 0.40-1.0  | —         | 12.0/14.0 | —        | —         | —        | —     | —     |            |
| 28.       | X85Cr18Mo2V          | 0.80/0.90                              | 1.0 Max   | 1.5 Max   | —         | 16.5/18.5 | 2.5 Max  | —         | —        | 0.035 | 0.045 | 0.60       |
| 29.       | X20Cr2Ni12N          | 0.15/0.25                              | 0.75-1.25 | 1.5 Max   | 10.5/12.5 | 20.0/22.0 | —        | —         | —        | 0.035 | 0.045 | Vanadium   |
| 30.       | X70Cr21Mn6Ni2N       | 0.65/0.75                              | 0.45-0.85 | 5.5-7.0   | 1.4/1.9   | 20.0/22.0 | —        | —         | —        | 0.035 | 0.045 | Max        |
| 31.       | X55Cr21Mn8Ni2N       | 0.50/0.60                              | 1.0 Max   | 7.0-9.5   | 1.5/2.75  | 20.0/22.0 | —        | —         | —        | 0.035 | 0.045 |            |
| 32.       | X53Cr22Mn9Ni4N       | 0.48/0.58                              | 0.25 Max  | 8.0-10.0  | 3.25/4.50 | 20.0/23.0 | —        | —         | —        | 0.035 | 0.045 |            |

\*For free cutting varieties at serial numbers 1 to 4, Sulphur and Selenium content shall be as agreed to between the purchaser and the manufacturer.

†For electrode steel Nb — 10 C to 1.0 in place of Ti.

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**TABLE 2 MECHANICAL TEST REQUIREMENTS IN ANNEALED  
QUENCHED OR SOLUTION TREATED CONDITION FOR  
PLATES/SHEETS/STRIPS**

( *Clauses 0.5 and 1.1* )

| STEEL DESIGNATION        | TENSILE    | 0.2 PERCENT          | ELONGATION     | HARDNESS, Max |                 |
|--------------------------|------------|----------------------|----------------|---------------|-----------------|
|                          | STRENGTH   | PROOF                | IN 50 mm       | Brinell       | Rock-<br>well B |
|                          | Min<br>MPa | STRESS<br>Min<br>MPa | Min<br>Percent |               |                 |
| Chromium — Nickel Steels |            |                      |                |               |                 |
| X02Cr19Ni10              | 485        | 170                  | 40             | 183           | 88              |
| X04Cr19Ni9               | 515        | 205                  | 40             | 183           | 88              |
| X07Cr18Ni9               | 515        | 205                  | 40             | 183           | 88              |
| X04Cr18Ni10Ti            | 515        | 205                  | 40             | 183           | 88              |
| X04Cr18Ni10Nb            | 515        | 205                  | 40             | 183           | 88              |
| X04Cr17Ni12Mo2           | 515        | 205                  | 40             | 217           | 95              |
| X02Cr17Ni12Mo2           | 485        | 170                  | 40             | 217           | 95              |
| X04Cr17Ni12Mo2Ti2        | 515        | 205                  | 40             | 217           | 95              |
| X04Cr19Ni13Mo3           | 515        | 205                  | 35             | 217           | 95              |
| X15Cr24Ni13              | 490        | 210                  | 40             | 223           | 95              |
| X20Cr25Ni20              | 515        | 210                  | 40             | 223           | 95              |
| X04Cr25Ni20              | 515        | 205                  | 40             | 217           | 95              |
| X07Cr17Mn12Ni4           | 550        | 250                  | 45             | 217           | 88              |
| Chromium — Steels        |            |                      |                |               |                 |
| X04Cr12                  | 415        | 205                  | 22.0           | 183           | 88              |
| X12Cr12                  | 450        | 205                  | 20.0           | 217           | 95              |
| X07Cr17                  | 450        | 205                  | 22.0           | 183           | 88              |
| X20Cr13                  | —          | —                    | —              | 241           | —               |
| X30Cr13                  | —          | —                    | —              | 241           | —               |
| X15Cr16Ni2               | —          | —                    | —              | 285           | —               |
| X108Cr17Mo               | —          | —                    | —              | 269           | —               |
| X15Cr25N                 | 515        | 275                  | 20             | 217           | —               |

**TABLE 3 MECHANICAL TEST REQUIREMENTS IN ANNEALED  
QUENCHED OR SOLUTION TREATED CONDITION FOR  
BARS AND FLATS**

( *Clauses 0.5 and 1.1* )

| STEEL DESIGNATION               | TENSILE<br>STRENGTH<br><i>Min</i><br>MPa | 0.2 PERCENT<br>PROOF STRESS<br><i>Min</i> ,<br>MPa | ELONGATION<br>IN 50 mm,<br><i>Min</i> ,<br>Percent | REDUCTION<br>OF AREA,<br><i>Min</i> ,<br>Percent | BRINELL<br>HARD-<br>NESS<br><i>Max</i> |
|---------------------------------|------------------------------------------|----------------------------------------------------|----------------------------------------------------|--------------------------------------------------|----------------------------------------|
| <b>Chromium — Nickel Steels</b> |                                          |                                                    |                                                    |                                                  |                                        |
| X02Cr19Ni10                     | 483                                      | 172                                                | 40                                                 | 50                                               | —                                      |
| X04Cr19Ni9                      | 517                                      | 207                                                | 40                                                 | 50                                               | —                                      |
| X10Cr18Ni9                      | 517                                      | 207                                                | 40                                                 | 50                                               | —                                      |
| X04Cr18Ni10Ti                   | 517                                      | 207                                                | 40                                                 | 50                                               | —                                      |
| X04Cr18Ni10Nb                   | 517                                      | 207                                                | 40                                                 | 50                                               | —                                      |
| X04Cr17Ni12Mo2                  | 517                                      | 207                                                | 40                                                 | 50                                               | —                                      |
| X02Cr17Ni12Mo2                  | 483                                      | 172                                                | 40                                                 | 50                                               | —                                      |
| X04Cr17Ni12Mo2Ti                | 517                                      | 207                                                | 40                                                 | 50                                               | —                                      |
| X04Cr19Ni13Mo3                  | 517                                      | 207                                                | 40                                                 | 50                                               | —                                      |
| X15Cr24Ni13                     | 490                                      | 210                                                | 40                                                 | 50                                               | —                                      |
| X20Cr25Ni20                     | 490                                      | 210                                                | 40                                                 | 50                                               | —                                      |
| X04Cr25Ni20                     | 517                                      | 207                                                | 40                                                 | 50                                               | —                                      |
| X0Cr17Mn6Ni4                    | 515                                      | 275                                                | 40                                                 | 45                                               | 217                                    |
| 40Ni14Cr14W3Si2                 | 785                                      | 345                                                | 35                                                 | 40                                               | 269                                    |
| <b>Chromium—Steels</b>          |                                          |                                                    |                                                    |                                                  |                                        |
| X04Cr12                         | 445                                      | 276                                                | 20                                                 | 45                                               | —                                      |
| X12Cr12                         | 483                                      | 276                                                | 20                                                 | 45                                               | —                                      |
| X07Cr17                         | 483                                      | 276                                                | 20                                                 | 45                                               | —                                      |
| X20Cr13                         | —                                        | —                                                  | —                                                  | —                                                | 241                                    |
| X30Cr13                         | —                                        | —                                                  | —                                                  | —                                                | 241                                    |
| X40Cr13                         | 600-750                                  | —                                                  | —                                                  | —                                                | 225                                    |
| X15Cr16Ni2                      | —                                        | —                                                  | —                                                  | —                                                | 285                                    |
| X108Cr17Mo                      | —                                        | —                                                  | —                                                  | —                                                | 269                                    |
| X15Cr25N                        | 490                                      | 280                                                | 16                                                 | 45                                               | 212                                    |

**TABLE 4 MECHANICAL PROPERTIES OF STAINLESS STEEL SHEETS, PLATES, STRIPS, BARS AND FLATS IN THE HARDENED AND TEMPERED CONDITION***( Clauses 0.5 and 1.1 )*

| STEEL DESIGNATION | 0.2 PERCENT<br>PROOF STRESS,<br>Min. MPa | TENSILE STRENGTH<br>MPa | ELONGATION<br>Min, Percent |
|-------------------|------------------------------------------|-------------------------|----------------------------|
| X12Cr12           | 410                                      | 590 to 780              | 16                         |
| X20Cr13           | 490                                      | 690 to 880              | 14                         |
| X30Cr13           | 590                                      | 780 to 980              | 11                         |
| X15Cr16Ni2        | 640                                      | 830 to 1 030            | 10                         |

NOTE — Elongation values are given on the Gauge Length as per IS : 3803-1976 that is,  $L_0 = 5.65\sqrt{S_0}$ . Percent elongation values given are applicable to all sizes of sheets, plates, strips and bars of 5 to 100 mm flats of 3 to 30 mm.

**TABLE 5 HARDNESS VALUES FOR CUTLERY STEELS IN THE FORM OF BARS, FLATS, SHEETS, PLATES AND STRIPS***( Clauses 0.5 and 1.1 )*

| STEEL DESIGNATION | QUENCHED AND TEMPERED<br>HARDNESS, Min |     |
|-------------------|----------------------------------------|-----|
|                   | HV                                     | HRC |
| X30Cr13           | 500                                    | 49  |
| X108Cr17Mo        | 660                                    | 58  |

NOTE — Annealed hardness of these grades has already been included in Tables 2 and 3.

**TABLE 6 HARDNESS VALUES FOR VALVE STEELS**

| STEEL DESIGNATION | HARDENED AND TEMPERED<br>HB |
|-------------------|-----------------------------|
| X45Cr9C13         | 255-293                     |
| X80Cr20Si2Ni1     | 255-306                     |
| X85Cr18Mo2V       | 255-306                     |
| X20Cr2Ni12N       | 302 <i>Max</i>              |
| X70Cr21Mn6Ni2N    | 321 <i>Min</i>              |
| X55Cr21Mn8Ni2N    | 321 <i>Min</i>              |
| X53Cr22Mn9Ni4N    | 321 <i>Min</i>              |

TABLE 7 RECOMMENDED HEAT TREATMENT FOR STAINLESS STEELS AND  
HEAT-RESISTING STEELS

( Clause 0.5 )

| STEEL DESIGNATION | ANNEALING/SOLUTION TREATMENT |             |                  | HARDENING AND TEMPERING TREATMENT |             |                  |                       |
|-------------------|------------------------------|-------------|------------------|-----------------------------------|-------------|------------------|-----------------------|
|                   | Symbol*                      | Temperature | Quenching Media* | Symbol*                           | Temperature | Quenching Media* | Tempering Temperature |
|                   | (1)                          | (2)         | (3)              | (5)                               | (6)         | (7)              | (8)                   |
|                   |                              | °C          |                  |                                   | °C          |                  | °C                    |
| X04Cr12           | A                            | 750-800     | f, a             | —                                 | —           | —                | —                     |
| X07Cr17           | A                            | 750-850     | a, w             | —                                 | —           | —                | —                     |
| X15Cr25N20        | A                            | 750-850     | a,               | —                                 | —           | —                | —                     |
| X12Cr12           | A                            | 700-780     | a                | Q+T                               | 950-1 000   | o, a             | 700-750               |
|                   | A                            | 770-870     | f                | —                                 | —           | —                | —                     |
| X20Cr13           | A                            | 770-870     | f                | Q+T                               | 980-1 030   | o, a             | 650-770               |
| X30Cr13           | A                            | 770-870     | f                | Q+T1                              | 980-1 030   | o, a             | 630-700               |
|                   |                              |             |                  | Q+T2                              | 980-1 030   | o, a             | 100-250               |
| X15Cr16Ni2        | A                            | 750-800     | a                | Q+T                               | 980-1 030   | o                | 630-700               |
| X108Cr17Mo        | A                            | 780-880     | f                | Q+T                               | 1 000-1 050 | o, a             | 100-250               |
| X02Cr19Ni10       | S                            | 1 000-1 150 | w, o, a          | —                                 | —           | —                | —                     |
| X04Cr19Ni8        | S                            | do          | do               | —                                 | —           | —                | —                     |
| X07Cr18Ni9        | S                            | do          | do               | —                                 | —           | —                | —                     |
| X04Cr18Ni10Ti     | S                            | do          | do               | —                                 | —           | —                | —                     |
| X04Cr18Ni10Nb     | S                            | do          | do               | —                                 | —           | —                | —                     |
| X04Cr17Ni12Mo2    | S                            | do          | do               | —                                 | —           | —                | —                     |
| X02Cr17Ni12Mo2    | S                            | do          | do               | —                                 | —           | —                | —                     |
| X04Cr17Ni12Mo2Ti2 | S                            | do          | do               | —                                 | —           | —                | —                     |
| X04Cr19Ni13Mo3    | S                            | do          | do               | —                                 | —           | —                | —                     |
| X07Cr17Mn12Ni4    | S                            | do          | do               | —                                 | —           | —                | —                     |

|                  |           |             |             |               |             |             |          |
|------------------|-----------|-------------|-------------|---------------|-------------|-------------|----------|
| X15Cr24Ni13      | <i>S</i>  | do          | do          | —             | —           | —           | —        |
| X20Cr25Ni20      | <i>S</i>  | do          | do          | —             | —           | —           | —        |
| X04Cr25Ni20      | <i>S</i>  | do          | do          | —             | —           | —           | —        |
| X85Cr18Mo2V      | —         | —           | —           | <i>Q+T</i>    | 1 050-1 080 | <i>o</i>    | 700-750  |
| X20Cr2Ni12N      | —         | —           | —           | <i>ST+(P)</i> | 1 100-1 200 | <i>o, w</i> | 700-800  |
| X70Cr21Mn6Ni2N   | —         | —           | —           | <i>ST+P</i>   | 1 100-1 200 | <i>w</i>    | 730-780† |
| X55Cr21Mn8Ni2N   | —         | —           | —           | <i>ST+P</i>   | 1 100-1 200 | <i>w</i>    | 730-780† |
| X53Cr22Mn9Ni4N   | —         | —           | —           | <i>ST+P</i>   | 1 100-1 200 | <i>w</i>    | 730-780† |
| X40Cr13          | <i>A</i>  | 750-850     | <i>F</i>    | —             | —           | —           | —        |
| X10Cr17Mn6Ni4    | <i>ST</i> | 1 000-1 150 | <i>w, o</i> | —             | —           | —           | —        |
| X40Ni14Cr14W3Si2 | <i>ST</i> | 980-1 080   | <i>w, o</i> | —             | —           | —           | —        |

NOTE — For austenitic stainless steels, the quenching media shall be selected taking into consideration the size of the material and carbon content of the grade.

\**A* = Annealing, *Q* = Quenching, *T* = Tempering

*S* = Softening, *f* = Furnace, *a* = Air, *w* = Water, *o* = Oil, *ST* = Solution Treatment  
*P* = Patenting

Stabilization Temperature 550°C

†Temperature for ageing treatment

## APPENDIX A

( Clause 0.8 )

## INFORMATION ON RULING SECTION

**A-1.** The mechanical properties obtained from a steel of a given composition are fundamentally influenced by the speed of cooling from a temperature above the upper critical point, that is, from hardening or normalizing temperature. Also for a given method of cooling, for example, oil quenching the rate of cooling is strongly affected by the size and shape of the part being treated. Therefore, for a steel of a given composition, the mechanical properties attained are affected not only by the method of cooling employed, but also by the size and shape of the part at the time of heat treatment. Thus, although it may be possible to obtain effective hardening by oil quenching a small section of a steel of a particular composition, it may be possible to satisfactory hardening throughout the mass when parts of larger cross-sectional dimensions are oil quenched. In such circumstances a steel of a different composition has to be selected. This inter-relation between the mechanical properties obtained and the size and shape of the part at the time of heat treatment is sometime referred to as 'mass effect'.

**A-2.** It is, therefore, necessary when selecting the steel which should be used to attain certain specified mechanical properties after heat treatment, to know the size and shape of the part to be heat-treated. The cross-sectional dimensions of that portion of the forging or part, where it is most important to obtain the desired mechanical properties, are regarded as the 'ruling section' of the part. In the case of round bar, the diameter is the ruling section, but as many shapes other than round bars require heat treatment, it is desirable to be able to relate the rates of cooling of other shapes to their equivalent sections of round bar. The diameters of round bars, the centres of which would cool through a given temperature range at the same rate as the centres of rectangular and square bars of selected sizes have been assessed and the resulting 'equivalent ruling sections' are given in Tables 8 and 9. Table 8 gives the equivalent ruling section in terms of diameter of round bar for rectangular and square sections when oil quenched, while Table 9 gives the same information for air cooling. As examples of how the tables may be applied, the centre of square section of 100 mm sides will, when oil quenched, cool at the same rate as a round section of 108 mm diameter and the equivalent ruling section of a slab 140 mm wide and 60 mm thick will, for oil quenching be 88 mm. In the case of other regular sections, close approximations may usually be made since the order of decreasing rate of cooling is round, octagonal, hexagonal and



square, while oval sections with major axis  $a$  and minor axis  $b$  will cool more slowly than a round bar of diameter  $b$  but faster than a rectangle  $axb$ .

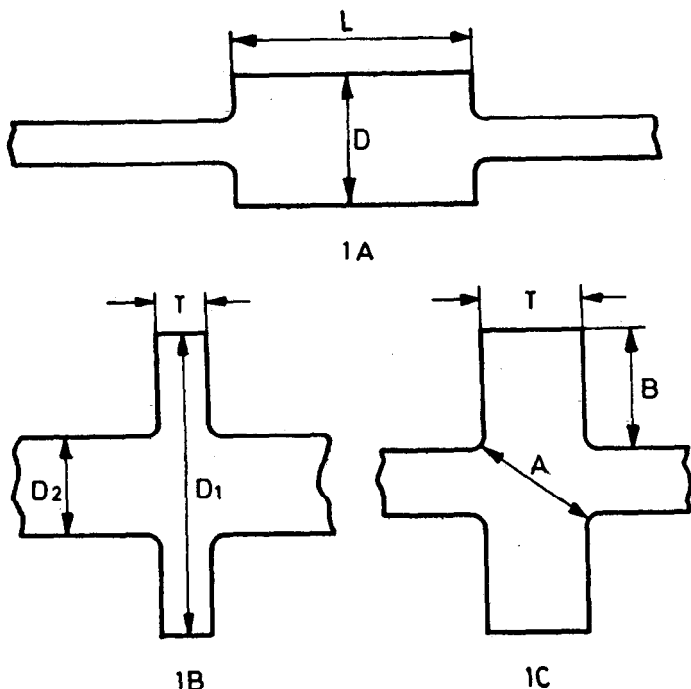


FIG. 1 RULING SECTION

A-3. In the case of parts of non-uniform section, not only it is necessary to consider which is the most important portion of the forging in which the specified properties are essential, but also to consider the relation of the length of any enlarged portion to its diameter. For example, in the case of a forging with a barrel, the length  $L$  of which is greater than the diameter  $D$  (see Fig. 1A), this diameter will be the ruling dimension. In the case of collar, the diameter  $D_1$  of which is greater than the thickness  $T$  (see Fig. 1B), the collar may be considered as a disc of which the thickness is the governing dimension although the ruling section of the part as a whole for practical purposes will be the diameter  $D_2$ . For forgings where the dimension  $B$  and thickness  $T$  of the portion of largest cross-section are similar (see Fig. 1C), the dimension  $A$ , between the centre and the nearest points of the external surface will in general determine the ruling section.

**TABLE 8 CONVERSION OF RECTANGULAR AND SQUARE SECTIONS  
INTO EQUIVALENT RULING SECTIONS, OIL QUENCHING**

( Clause A-2 )

All dimensions in millimetres.

| THICKNESS →<br><br>WIDTH ↓ | 10                            | 20  | 30  | 40  | 50  | 60  | 80  | 100 | 120  | 140  |
|----------------------------|-------------------------------|-----|-----|-----|-----|-----|-----|-----|------|------|
|                            | DIAMETER OF EQUIVALENT ROUNDS |     |     |     |     |     |     |     |      |      |
| (1)                        | (2)                           | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) |
| 10                         | 10                            | —   | —   | —   | —   | —   | —   | —   | —    | —    |
| 20                         | 14                            | 21  | —   | —   | —   | —   | —   | —   | —    | —    |
| 30                         | 16                            | 26  | 32  | —   | —   | —   | —   | —   | —    | —    |
| 40                         | 17                            | 29  | 37  | 43  | —   | —   | —   | —   | —    | —    |
| 50                         | 17                            | 31  | 40  | 48  | 54  | —   | —   | —   | —    | —    |
| 60                         | 17                            | 32  | 43  | 51  | 59  | 65  | —   | —   | —    | —    |
| 80                         | 17                            | 32  | 46  | 57  | 66  | 74  | 87  | —   | —    | —    |
| 100                        | 17                            | 33  | 47  | 60  | 71  | 80  | 96  | 108 | —    | —    |
| 120                        | 17                            | 33  | 48  | 62  | 74  | 85  | 102 | 117 | 130  | —    |
| 140                        | 17                            | 33  | 48  | 63  | 76  | 88  | 108 | 125 | 140  | 152  |
| 160                        | 17                            | 33  | 48  | 63  | 77  | 90  | 112 | 131 | 148  | X    |
| 180                        | 17                            | 33  | 48  | 63  | 77  | 91  | 115 | 136 | 154  | X    |
| 200                        | 17                            | 33  | 48  | 63  | 78  | 92  | 117 | 140 | X    | X    |
| 225                        | 17                            | 33  | 48  | 63  | 78  | 92  | 119 | 143 | X    | X    |
| 250                        | 17                            | 33  | 48  | 63  | 78  | 92  | 120 | 145 | X    | X    |
| 300                        | 17                            | 33  | 48  | 63  | 78  | 92  | 121 | 148 | X    | X    |
| 350 and over               | 18                            | 33  | 49  | 63  | 78  | 92  | 121 | 150 | X    | X    |

X = Greater than 160 mm.

**TABLE 9 CONVERSION OF RECTANGULAR AND SQUARE SECTIONS INTO EQUIVALENT RULING SECTIONS, AIR COOLING**

( Clause A-2 )

All dimensions in millimetres.

| THICKNESS→<br>WIDTH↓ | 10                            | 20  | 30  | 40  | 50  | 60  | 80  | 100 | 120  | 140  |
|----------------------|-------------------------------|-----|-----|-----|-----|-----|-----|-----|------|------|
|                      | DIAMETER OF EQUIVALENT ROUNDS |     |     |     |     |     |     |     |      |      |
| (1)                  | (2)                           | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) |
| 10                   | 10                            | —   | —   | —   | —   | —   | —   | —   | —    | —    |
| 20                   | 13                            | 20  | —   | —   | —   | —   | —   | —   | —    | —    |
| 30                   | 15                            | 24  | 30  | —   | —   | —   | —   | —   | —    | —    |
| 40                   | 16                            | 27  | 34  | 40  | —   | —   | —   | —   | —    | —    |
| 50                   | 16                            | 29  | 38  | 45  | 51  | —   | —   | —   | —    | —    |
| 60                   | 17                            | 30  | 41  | 49  | 55  | 61  | —   | —   | —    | —    |
| 80                   | 17                            | 34  | 45  | 54  | 62  | 70  | 81  | —   | —    | —    |
| 100                  | 17                            | 33  | 47  | 58  | 68  | 76  | 90  | 102 | —    | —    |
| 120                  | 17                            | 34  | 49  | 62  | 72  | 82  | 98  | 111 | 122  | —    |
| 140                  | 17                            | 34  | 50  | 64  | 76  | 86  | 104 | 119 | 132  | 142  |
| 160                  | 17                            | 34  | 51  | 66  | 79  | 90  | 109 | 126 | 140  | 152  |
| 180                  | 18                            | 35  | 51  | 67  | 81  | 93  | 114 | 132 | 148  | X    |
| 200                  | 18                            | 35  | 52  | 68  | 83  | 95  | 117 | 137 | 154  | X    |
| 225                  | 18                            | 35  | 52  | 69  | 85  | 98  | 121 | 142 | X    | X    |
| 250                  | 18                            | 35  | 53  | 70  | 86  | 100 | 125 | 147 | X    | X    |
| 300                  | 18                            | 37  | 54  | 71  | 88  | 104 | 131 | 155 | X    | X    |
| 350 and over         | 20                            | 39  | 59  | 78  | 97  | 115 | 152 | ×   | X    | X    |

X = Greater than 160 mm.

( Continued from page 2 )

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